particle having definite spin values. By using axes at three angles, Bell provided a way to count Pauli's angels.

No Smoke but Fire

In case you missed any of the details, let's summarize where we've gotten. Through the Heisenberg uncertainty principle, quantum mechanics claims that there are features of the world—like the position and the velocity of a particle, or the spin of a particle about various axes—that cannot simultaneously have definite values. *A particle, according to quantum theory, cannot have a definite position and a definite velocity; a particle cannot have a definite spin (clockwise or counterclockwise) about more than one axis; a particle cannot simultaneously have definite attributes for things that lie on opposite sides of the uncertainty divide.* Instead, particles hover in quantum limbo, in a fuzzy, amorphous, probabilistic mixture of all possibilities; only when measured is one definite outcome selected from the many. Clearly, this is a drastically different picture of reality than that painted by classical physics.

Ever the skeptic about quantum mechanics, Einstein, together with his colleagues Podolsky and Rosen, tried to use this aspect of quantum mechanics as a weapon against the theory itself. EPR argued that even though quantum mechanics does not allow such features to be simultaneously determined, particles nevertheless do have definite values for position and velocity; particles do have definite spin values about all axes; particles do have definite values for all things forbidden by quantum uncertainty. EPR thus argued that quantum mechanics cannot handle all elements of physical reality—it cannot handle the position and velocity of a particle; it cannot handle the spin of a particle about more than one axis—and hence is an incomplete theory.

For a long time, the issue of whether EPR were correct seemed more a question of metaphysics than of physics. As Pauli said, if you can't actually measure features forbidden by quantum uncertainty, what difference could it possibly make if they, nevertheless, exist in some hidden fold of reality? But, remarkably, John Bell found something that had escaped Einstein, Bohr, and all the other giants of twentieth-century